KEN KANGYI PENG

(778) · 952 · 9839 ◇ kpeng2@uottawa.ca ◇ https://kenkpeng.github.io/

Curriculum Vitae

EDUCATION

PhD in Statistics *Sep 2025*
Simon Fraser University, BC, Canada
Thesis: "Characterizing Stochastic Processes with Applications in Infectious Disease and Sports Analytics"
Supervisors: X. Joan Hu and Tim Swartz

MSc in Statistics *May 2022*
Simon Fraser University, BC, Canada
Thesis: "Bayesian Approaches for Critical Velocity Models"
Supervisors: Tim Swartz and Gary Parker

BSc in Statistics & Minor in Mathematics *May 2020*
Simon Fraser University, BC, Canada

RESEARCH POSITIONS

Research

- **CANSSI Distinguished Postdoctoral Fellow (incoming)** *Jan 2027 (planned start)*
  Simon Fraser University and York University
  Project: AI-Powered Statistical Analysis for Elite Team Sports
  Supervision: Tim Swartz and Tianyu Guan
- **Postdoctoral Fellow** *Sep 2025 – Dec 2026*
  University of Ottawa, Department of Civil Engineering
  Project: The Application of Statistical Methods to Wastewater Analysis
  Supervision: Robert Delatolla and Charmaine Dean
- **Research Assistant** *May 2021 – Sep 2025*
  Simon Fraser University, Department of Statistics and Actuarial Science, Sports Analytics Group
  Project: CANSSI CRT "Sports Analytics" & CANSSI CRT "The Application of Statistical Methods to Wastewater Analysis"
  Supervision: Tim Swartz and X. Joan Hu
- **Research Assistant** *Jan 2024 – Sep 2024*
  University of Ottawa, Department of Civil Engineering
  Project: CoVaRR-Net Wastewater Surveillance Rapid Response Research Group
  Supervision: Robert Delatolla
- **Research Assistant** *Jan 2022 – May 2023*
  University of Waterloo, Department of Statistics and Actuarial Science
  Projects: MfPH - NSERC "Emerging Infectious Diseases Modelling Initiative"
  Supervision: Charmaine Dean
- **Research Assistant** *Mar 2020 – May 2021*
  Simon Fraser University, Department of Biomedical Physiology and Kinesiology
  Supervision: Tim Swartz and David C. Clarke

Industry and Application Research Experience

- **Co-founder & Chief Data Scientist** *Jun 2023 – Current*
  AccMov Health Inc.
  Statistical and data science consulting for sports and health research
- **Sports Analyst** *Jan 2022 – Dec 2023*
  Stathletes
  Hockey analytics
- **Co-op Data Scientist** *Oct 2021 – Dec 2021*
  Public Health Agency of Canada
  Worked with the ArriveCAN app database
- **Co-op Statistician** *May 2019 – Sep 2019*
  BC Centre for Excellence in HIV/AIDS
  Worked with health administrative data
- **Co-op Data Scientist** *May 2018 – Dec 2018*
  Statistics Canada
  Worked with Canada's multifactor productivity database & T2-LEAP linked administrative database

RESEARCH PAPERS

Refereed Papers

1. **Peng, K. K.**, Delatolla, R., Hu, X. J., Renouf, E., & Dean, C. B. "Joint Modeling of Two Stochastic Processes, with Application to Learning Hospitalization Dynamics from Wastewater Viral Concentrations." *Canadian Journal of Statistics*, in press. arXiv:2601.07977.
2. Wen, J., **Peng, K. K.**, Lee, B. E., et al. "Site-specific wastewater-based surveillance in early detection of COVID-19 new cases and prediction of mass testing outcomes in long-term care facilities." *Scientific Reports*, 16, 17235 (2026). doi:10.1038/s41598-026-47833-8.
3. Hegazy, N., **Peng, K. K.**, de Haan-Ward, J., et al. "Twelve distinct laboratory methods used to measure SARS-CoV-2 in wastewaters throughout a three-year Ontario-wide, Canada study: Impact on public health interpretation of disease incidence." *Journal of Water Process Engineering*, 89, 110265 (2026). doi:10.1016/j.jwpe.2026.110265.
4. Isaacs, R. L.*, Hu, X. J., **Peng, K. K.**, & Swartz, T. B. "Learning About Corner Kicks in Soccer by Analysis of Event Times Using a Frailty Model." In *Sports Analytics (ISACE 2026)*, Lecture Notes in Computer Science, 16610, 442–455 (2026). doi:10.1007/978-3-032-27272-0_28.
5. Zhao, X., **Peng, K. K.**, Cournoyer, J., & Graham, R. B. "Seasonal changes in neuromuscular and biomechanical performance in female soccer players: Implications for athlete monitoring." *International Journal of Sports Science & Coaching*, OnlineFirst (2026). doi:10.1177/17479541261479187.
6. **Peng, K. K.**, Dean, C. B., Hu, X. J., & Delatolla, R. "Learning associations of COVID-19 hospitalizations with wastewater viral signals by Markov modulated models." *Epidemics*, 52, 100840 (2025). doi:10.1016/j.epidem.2025.100840.
7. **Peng, K. K.**, Hu, X. J., & Swartz, T. B. "On the time of corner kicks in soccer: an analysis of event history data." *Computational Statistics*, 40, 2067–2083 (2025; online 2024). doi:10.1007/s00180-024-01567-1.
8. Hegazy, N., **Peng, K. K.**, D'Aoust, P. M., et al. "Variability of Clinical Metrics in Small Population Communities Drive Perceived Wastewater and Environmental Surveillance Data Quality: Ontario, Canada-Wide Study." *ACS ES&T Water*, 5(4), 1605–1619 (2025). doi:10.1021/acsestwater.4c00958.
9. **Peng, K. K.**, Brodie, R. T., Swartz, T. B., & Clarke, D. C. "Bayesian inference of the impulse-response model of athlete training and performance." *International Journal of Performance Analysis in Sport*, 24(1), 74–89 (2024; online 2023). doi:10.1080/24748668.2023.2268480.
10. **Peng, K. K.**, Renouf, E. M., Dean, C. B., Hu, X. J., Delatolla, R., & Manuel, D. G. "An exploration of the relationship between wastewater viral signals and COVID-19 hospitalizations in Ottawa, Canada." *Infectious Disease Modelling*, 8(3), 617–631 (2023). doi:10.1016/j.idm.2023.05.011.

11. **Peng, K. K.**, Clarke, D. C., & Swartz, T. B. "Bayesian approaches for critical velocity modelling of data from intermittent efforts." *International Journal of Sports Science & Coaching*, 17(4), 868–879 (2022). doi:10.1177/17479541221100311.

*Mentored student.

Submitted Manuscripts

12. **Peng, K. K.**, Hu, X. J., & Swartz, T. B. "Modeling Event Dynamics by Self-Exciting Processes with Random Memory." Submitted. arXiv:2601.07980.
13. **Peng, K. K.**, Guan, T., & Zhao, X. "Hierarchical Bayesian Joint Modeling of Athlete Performance Trajectories and Latent Peak Ages." Submitted. arXiv:2610.35473.
14. **Peng, K. K.**, Xian, C., Dean, C. B., Renouf, E. M., Hu, X. J., & Delatolla, R. "Bayesian Spatio-temporal Modeling of Multi-Site Wastewater Data with Time Shifts." Submitted.

TEACHING EXPERIENCE

- **Instructor** *Spring 2026*
  Simon Fraser University
  STAT 445/645: Applied Multivariate Analysis
  – Taught a combined upper-level undergraduate and graduate course.
  – Designed all examinations, assignments, and tutorial materials, aligning them with lectures and integrating analytical work with implementation in `R`.
- **Guest Lecturer**
  Simon Fraser University
  STAT 240: Lecture on text data manipulation *Spring 2025*
- **Teaching Assistant**
  Simon Fraser University
  STAT 240: Introduction to Data Science *Spring 2025*
  STAT 302: Analysis of Experimental and Observational Data *Fall 2023*
  STAT 830: Statistical Theory I *Fall 2022*
  STAT 261: Laboratory for Introductory R for Data Science *Fall 2021*
  STAT 201: Statistics for the Life Sciences *Summer 2021*
  STAT 380: Introduction to Stochastic Processes *Spring 2021*
  STAT 203: Introduction to Statistics for the Social Sciences *Fall 2020*
  STAT 100: Chance and Data Analysis *Fall 2020*

Mentoring

- **Shilin Mao** *May 2026 – Present*
  Undergraduate student in Computer Science, Simon Fraser University
  Project: *Simulating Temporal Dynamics of Soccer Game*
  Undergraduate Student Research Award (USRA), Summer 2026 – $6,000 CAD
  Mentoring with X. Joan Hu
- **Julian Lau** *Aug 2025 – Present*
  BSc (Honours) student in Physiology, University of Alberta
  Project: *Evaluation of Wastewater-Based Surveillance Decision Rules for RSV Season Timing in a Northern Alberta Pediatric Prevention Program*
  Mentoring with Bonita Lee and Lilly Pang
- **Krystal He** *May 2025 – Present*
  MSc student in Statistics, Simon Fraser University
  Project: Spatio-temporal analysis of COVID-19 wastewater surveillance data in the Netherlands
  Mentoring with X. Joan Hu

- **Riley L. Isaacs** *May 2024 – Present*
  MSc student in Statistics, Simon Fraser University
  Project: *Multilevel Frailty Modeling of Soccer Corner-Kick Timing*
  Mentoring with X. Joan Hu; resulting refereed paper listed above*

HONORS AND AWARDS

- CANSSI Distinguished Postdoctoral Fellowship – $140,000 CAD *2026–2028 cohort*
  Awarded; deferred start: January 2027
- Multiple Graduate Fellowships, SFU – Total: $17,500 CAD *2021–2025*
- Multiple PhD Research Scholarships, SFU – Total: $9,122 CAD *2023–2025*
- SFU Travel and Research Award – $2,054 CAD *2024–2025*
- MfPH – Career Networking & Industry Partnership Event Travel Funding – $1,500 CAD *Fall 2023*
- Mathematics for Public Health Festival Travel Funding – $2,000 CAD *Fall 2022*
- Dr. Gord Sleivert Young Investigator Award – $500 CAD *Summer 2020*
- NSERC Undergraduate Student Research Award – $4,500 CAD *Summer 2019*

SOFTWARE DEVELOPMENT

- **Myotion** — Open-source software for electromyography signal processing, visualization, and statistical analysis.
  Chen, G., Zhang, Y., Zhao, X., **Peng, K. K.**, Ma, Y., and Yang, C. (2026).
  Repository: `https://github.com/AccMov/Myotion_Open`.
  Project DOI: 10.17605/OSF.IO/PRKET.
- **peakAge** — R package for Bayesian joint modeling of athlete performance trajectories and latent peak ages.
  Author and maintainer. Supports symmetric and asymmetric performance curves, covariate effects, and uncertainty quantification. *In development.*
  Repository: `https://github.com/KenKPeng/peakAge`.

PRESENTATIONS

Invited Presentations

- 2026 (Upcoming) 20th International Joint Conference CFE-CMStatistics: Recent advancements in biostatistical methods
  Talk: *Spatio-temporal covariance modeling for multi-site wastewater surveillance*
  Berlin, Germany
- 2026 (Upcoming) BIRS workshop (26w5507): Advances in Statistical Methodology for the Analysis of Wastewater Data
  Talk: *Benchmark-Based Statistical Assessment of Inter-Laboratory Variability in Wastewater SARS-CoV-2 Measurements*
  Banff, AB, Canada
- 2026 The 7th ICSA-Canada Chapter Symposium: Applications of functional data, time series data, and stochastic processes
  Talk: *Spatio-Temporal Covariance Modeling for Multi-Site Wastewater Surveillance*
  Montreal, QC, Canada
- 2026 Conference on Econometrics and Statistics: Sports analytics (Special Invited Session)
  Talk: *Hierarchical Bayesian Inference of Athlete Performance Trajectories and Latent Peak Ages*
  Kyoto, Japan

- 2026 Centre de recherches mathématiques (CRM): Invited presentation (joint with Professor Charmaine Dean)
  Talk: *Statistical Learning from Wastewater Surveillance: Latent Infection Dynamics, Aggregated Data, and Hospitalization Burden*
  Montreal, QC, Canada
- 2026 WNAR of the International Biometric Society: Novel statistical methods for complex event history data
  Talk: *Modeling Event Dynamics by Self-Exciting Processes with Random Memory*
  Pullman, Washington, US
- 2026 Department of Mathematics and Statistics Seminar, Wuhan University
  Talk: *Statistical Challenges in Wastewater-Based Infectious Disease Surveillance*
  Wuhan, China
- 2025 Conference on Econometrics and Statistics: Infectious disease modeling
  Talk: *Joint Modeling of Wastewater Viral Loads (WWS) and Hospitalization Dynamic*
  Tokyo, Japan
- 2025 International Conference on Statistics and Data Science: Environmental modelling
  Talk: *Exploring Associations of Risk of COVID-19 Hospitalizations with Wastewater Viral Signals by Markov Modulated Model with Distributed Lasting Time*
  Vancouver, BC, Canada
- 2025 WNAR of the International Biometric Society: Novel models of survival data
  Talk: *Modeling Event Dynamics by Self-Exciting Processes with Random Memory*
  Whistler, BC, Canada
- 2025 Lifetime Data Science: Recent advances in life history data analysis with multi-state models
  Talk: *Joint Modeling of Wastewater Viral Loads (WWS) and Hospitalization Dynamics*
  Brooklyn, NY, US
- 2024 Joint Statistical Meetings: Newly developed point process models for repeated events
  Talk: *A Novel Stochastic Process Model For Exploring Occurrences of Corner Kicks In Soccer*
  Portland, Oregon, US
- 2024 WNAR of the International Biometric Society: Sparse modeling for biomedical data
  Talk: *Exploring associations of risk of COVID-19 hospitalizations with wastewater viral signals by Markov modulated model with distributed lasting time*
  Fort Collins, Colorado, US
- 2023 WNAR of The International Biometric Society: Statistics in biosciences
  Talk: *How can we use wastewater viral signals to learn about COVID-19?*
  Anchorage, Alaska, US
- 2022 BIRS-CMO: Statistical challenges in the identification, validation, and use of surrogate markers
  Talk: *The Association between COVID-19 Hospitalizations and SARS-CoV-2 Wastewater Viral Signals in Ottawa, Ontario*
  Banff International Research Station. Remote

Other Presentations

- 2026 iMicroSeq meeting: Methods evaluation to enable future data sharing and integration
  Remote
- 2023 Fields MfPH - Career Networking and Industry Partnership Event, The Fields Institute
  Toronto, ON, Canada
- 2023 North America Machine Learning, Optimization and Statistics Symposium, poster presentation
  Vancouver, BC, Canada
- 2022–2023 Colloquium on Mathematics for Public Health, The Fields Institute
  Remote
- 2022 Mathematics for Public Health Festival (MfPHest), Poster presentation, The Fields Institute
  Toronto, ON, Canada

- 2021 Canadian Statistical Sciences Institute collaborative research teams showcase event
  Remote
- 2021 Statistical Society of Canada annual meeting, Poster presentation
  Remote
- 2020 The SPort INnovation Summit, recipient of the Dr. Gord Sleivert Young Investigator Award
  Remote

ACADEMIC ACTIVITIES

Conference and Workshop Service

- **Organizer**, Accepted invited session at the 2027 Lifetime Data Science Conference: *New Perspectives on Lifetime Data Science through Sports* *2027*
- **Organizer**, Accepted invited session at the 2027 Statistical Society of Canada Annual Meeting: *Advances in Sports Analytics* *2027*
- **Scientific Program Committee Member**, EcoSta 2027 *2027*
- **Chair**, Invited session at 2026 ICSA: *Recent Statistical Developments Stimulated by COVID-19 Pandemic* *2026*
- **Organizer & Chair**, Invited session at 2026 EcoSta: *Statistical modeling for environmental surveillance* *2026*
- **Co-organizer**, BIRS 5-day workshop: *Advances in Statistical Methodology for the Analysis of Wastewater Data* *2026*
- **Review Committee Member**, 2026 WNAR Student Paper Competition *2026*
- **Co-organizer & Chair**, Invited session at 2025 EcoSta: *Innovative statistical approaches in wastewater-based epidemiology: Recent developments and applications* *2025*
- **Volunteer**, International Conference on Statistics and Data Science, Vancouver *2025*
- **Program Assistant**, Cascadia Symposium on Statistics in Sports, Vancouver *2023–2024*
- **Volunteer**, North America Machine Learning, Optimization and Statistics Symposium, Vancouver *Summer 2023*

Journal Peer Review

- Statistics in Medicine
- Computational Statistics
- Canadian Journal of Statistics
- Statistical Theory and Related Fields
- BMC Medical Research Methodology
- Journal of Hospital Infection

Academic Service

- **Committee Member**, Student and Recent Graduate Committee (SARGC), Statistical Society of Canada *2026–2027*
- **Lab Meeting Coordinator**, Dr. Joan Hu's Lab *2024–2025*
- **Next Generation Member**, The Fields Institute for Research in Mathematical Sciences, Mathematics for Public Health (MfPH) program *2022–2024*